REVIEW PACKAGE — STOP BEFORE MANUFACTURING EXPORTS

USB-powered Teensy / six socketed Minis / through-hole assembly / fused 12V motor power

Identical global labels are the same connection. All components are passive carrier connections.

Drill-verified Mini v1.0 footprint; encoder contact 1 and 160 x 120 mm base retained. Review before ordering.

01 / USB-powered Teensy sockets

File: controller.kicad_sch

02 / A1: index_curl_1

File: axis1.kicad_sch

03 / A2: index_curl_2

File: axis2.kicad_sch

04 / A3: index_splay

File: axis3.kicad_sch

05 / A4: middle_curl_1

File: axis4.kicad_sch

06 / A5: middle_curl_2

File: axis5.kicad_sch

07 / A6: middle_splay

File: axis6.kicad_sch

08 / Fused 12V and ground star

File: power.kicad_sch

Sheet: /
File: trueskate-main-v1.kicad_sch

Title: TrueSkate USB-powered six-axis carrier

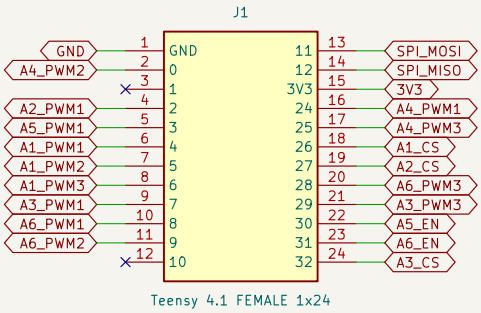
Size: A3	Date:	**Rev: v1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 1/9

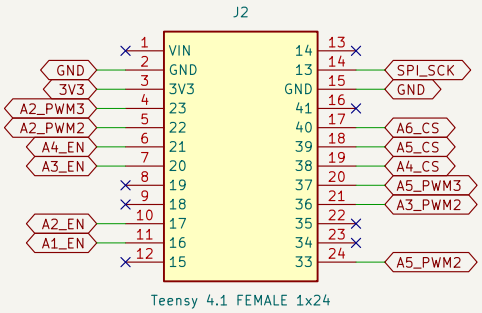
USB is the ONLY Teensy power source. VIN socket contact is deliberately unconnected.

All six axes use edge-aligned PWM. Existing firmware needs the new EN/CS assignments.

Teensy 3.3V powers encoders. Mini regulator outputs are NOT connected to this rail.

Mini v1.0 drill-derived footprint; encoder pin 1 and base dimensions retained.





Sheet: /01 / USB-powered Teensy sockets/
File: controller.kicad_sch

Title: 01 / USB-powered Teensy sockets

Size: A4	Date:	Rev: v1 REVIEW
KiCad E.D.A. 10.0.6		Id: 2/9

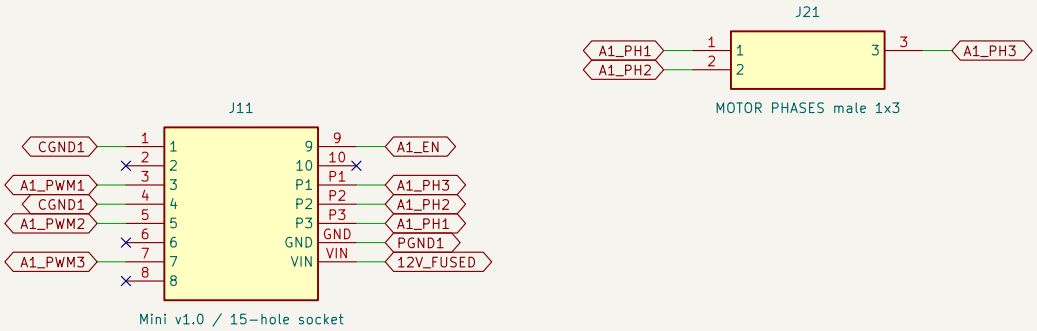
Mini v1.0: exact 15-hole footprint; component side up, pins down; NOT mirrored.

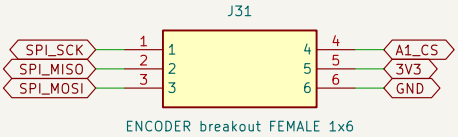
EN goes directly to Teensy; 10k to this Mini's power ground holds EN low at reset.

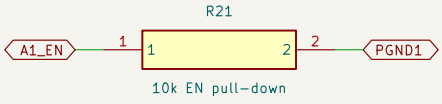
JP control-ground bridge defaults OPEN; module references its directly routed power-return pad.

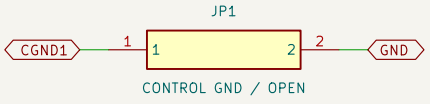
Breakout socket: 1 CLK, 2 MISO, 3 MOSI, 4 CSn, 5 3.3V, 6 GND. Pin 1 is marked.

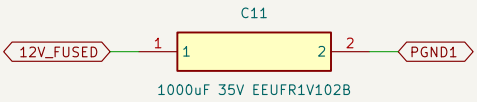
Mini 3.3V output, nRESET, nSLEEP and nFAULT contacts intentionally unconnected.

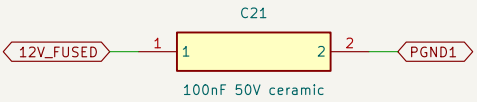












Sheet: /02 / A1: index_curl_1/
File: axis1.kicad_sch

Title: 02 / A1: index_curl_1

Size: A4	Date:	Rev: v1 REVIEW
KiCad E.D.A. 10.0.6		Id: 3/9

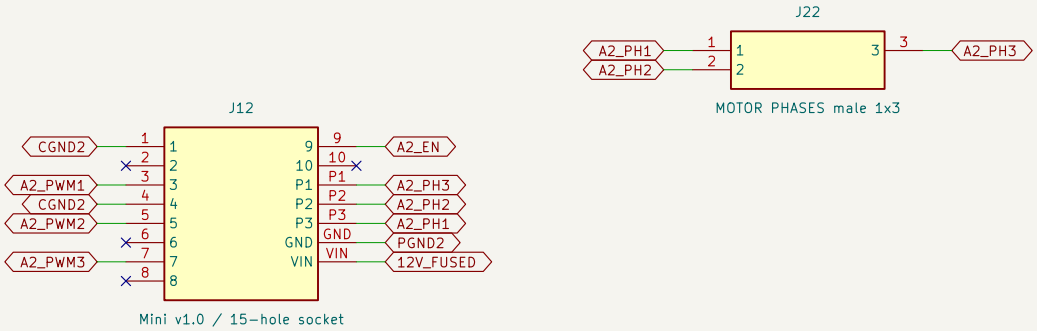
Mini v1.0: exact 15-hole footprint; component side up, pins down; NOT mirrored.

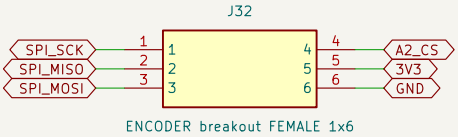
EN goes directly to Teensy; 10k to this Mini's power ground holds EN low at reset.

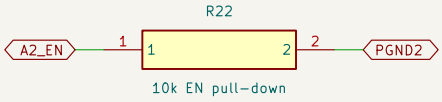
JP control-ground bridge defaults OPEN; module references its directly routed power-return pad.

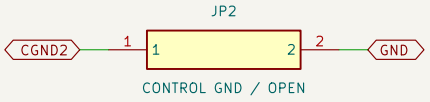
Breakout socket: 1 CLK, 2 MISO, 3 MOSI, 4 CSn, 5 3.3V, 6 GND. Pin 1 is marked.

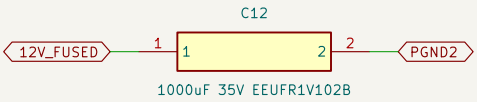
Mini 3.3V output, nRESET, nSLEEP and nFAULT contacts intentionally unconnected.

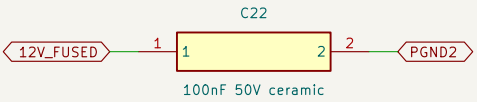












Sheet: /03 / A2: index_curl_2/
File: axis2.kicad_sch

Title: 03 / A2: index_curl_2

Size: A4	Date:	Rev: v1 REVIEW
KiCad E.D.A. 10.0.6		Id: 4/9

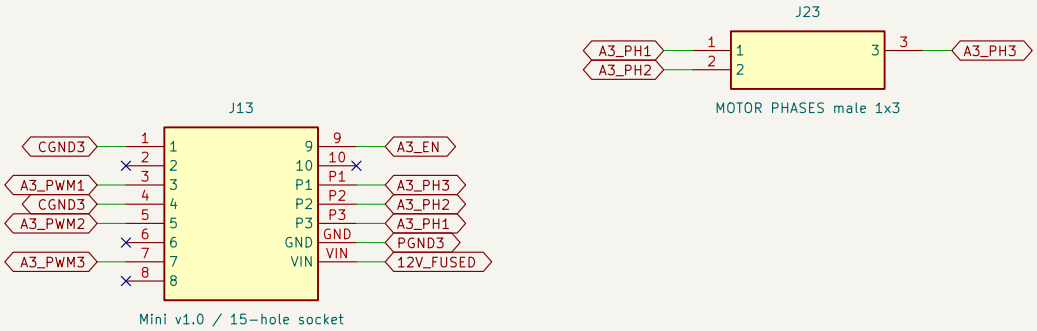
Mini v1.0: exact 15-hole footprint; component side up, pins down; NOT mirrored.

EN goes directly to Teensy; 10k to this Mini's power ground holds EN low at reset.

JP control-ground bridge defaults OPEN; module references its directly routed power-return pad.

Breakout socket: 1 CLK, 2 MISO, 3 MOSI, 4 CSn, 5 3.3V, 6 GND. Pin 1 is marked.

Mini 3.3V output, nRESET, nSLEEP and nFAULT contacts intentionally unconnected.



Sheet: /04 / A3: index_splay/		
File: axis3.kicad_sch		
Title: 04 / A3: index_splay		
Size: A4	Date:	**Rev: v1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 5/9

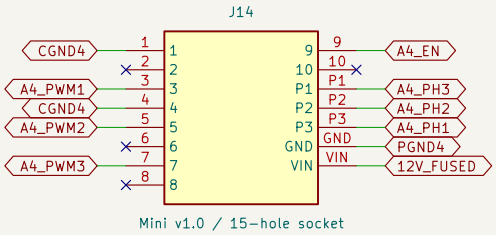
Mini v1.0: exact 15-hole footprint; component side up, pins down; NOT mirrored.

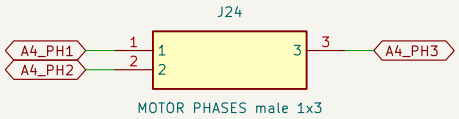
EN goes directly to Teensy; 10k to this Mini's power ground holds EN low at reset.

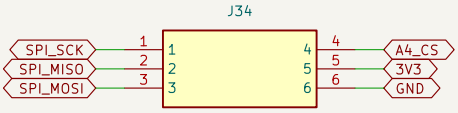
JP control-ground bridge defaults OPEN; module references its directly routed power-return pad.

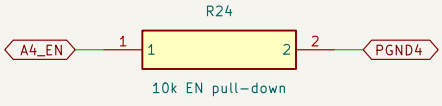
Breakout socket: 1 CLK, 2 MISO, 3 MOSI, 4 CSn, 5 3.3V, 6 GND. Pin 1 is marked.

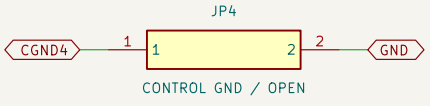
Mini 3.3V output, nRESET, nSLEEP and nFAULT contacts intentionally unconnected.

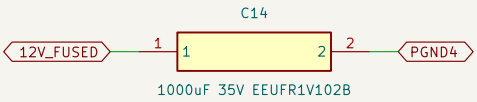


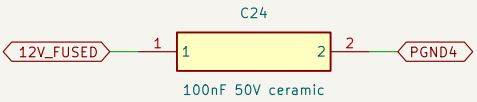












Sheet: /05 / A4: middle_curl_1/
File: axis4.kicad_sch

Title: 05 / A4: middle_curl_1

Size: A4	Date:	Rev: v1 REVIEW
KiCad E.D.A. 10.0.6		Id: 6/9

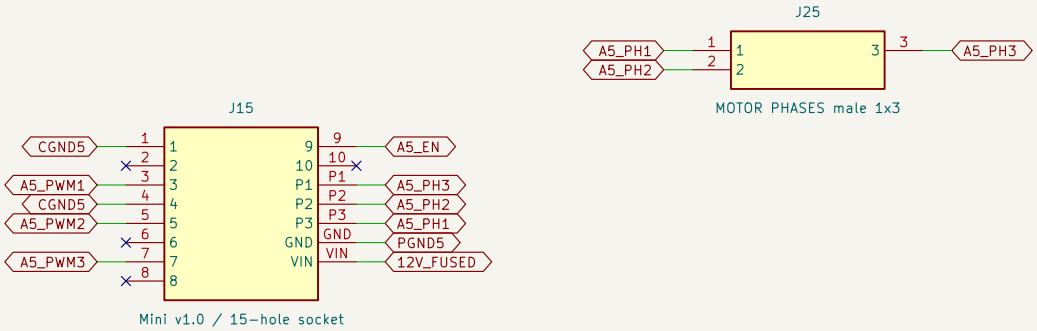
Mini v1.0: exact 15-hole footprint; component side up, pins down; NOT mirrored.

EN goes directly to Teensy; 10k to this Mini's power ground holds EN low at reset.

JP control-ground bridge defaults OPEN; module references its directly routed power-return pad.

Breakout socket: 1 CLK, 2 MISO, 3 MOSI, 4 CSn, 5 3.3V, 6 GND. Pin 1 is marked.

Mini 3.3V output, nRESET, nSLEEP and nFAULT contacts intentionally unconnected.



Sheet: /06 / A5: middle_curl_2/
File: axis5.kicad_sch

Title: 06 / A5: middle_curl_2

Size: A4	Date:	Rev: v1 REVIEW
KiCad E.D.A. 10.0.6		Id: 7/9

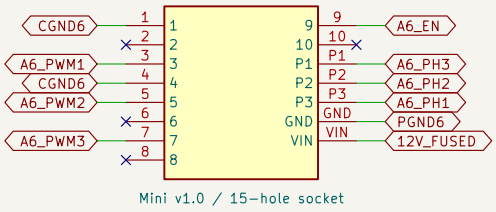
Mini v1.0: exact 15-hole footprint; component side up, pins down; NOT mirrored.

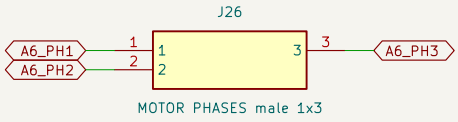
EN goes directly to Teensy; 10k to this Mini's power ground holds EN low at reset.

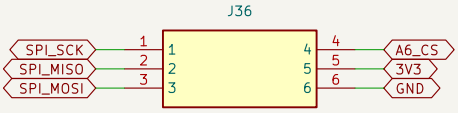
JP control-ground bridge defaults OPEN; module references its directly routed power-return pad.

Breakout socket: 1 CLK, 2 MISO, 3 MOSI, 4 CSn, 5 3.3V, 6 GND. Pin 1 is marked.

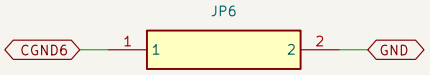
Mini 3.3V output, nRESET, nSLEEP and nFAULT contacts intentionally unconnected.

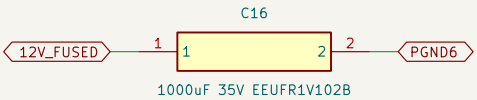


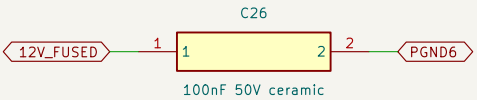












Sheet: /07 / A6: middle_splay/ File: axis6.kicad_sch		
Title: 07 / A6: middle_splay		
Size: A4	Date:	**Rev: v1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 8/9

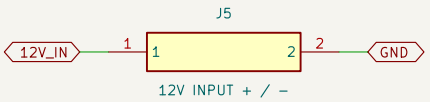
12V is motor power only: no connection to Teensy VIN, VUSB or 3.3V.

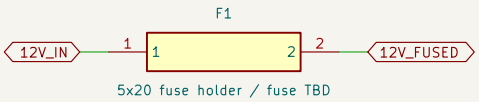
Through-hole fuse holder; candidate 3.15A time-delay cartridge, qualify load/inrush.

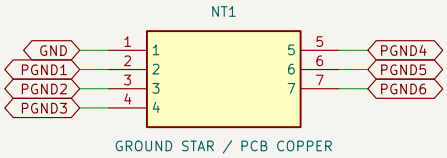
Correct input polarity required. Reverse-polarity protection moved to the v2 backlog.

NT1 is PCB copper: six separate motor returns and logic ground meet at this junction.

No hardware kill on v1. Disconnect 12V to remove motor power.







Sheet: /08 / Fused 12V and ground star/
File: power.kicad_sch

Title: 08 / Fused 12V and ground star

Size: A4	Date:	Rev: v1 REVIEW
KiCad E.D.A. 10.0.6		Id: 9/9